BYTE NDT · 🇬🇧 English · P03 / Manual edition 1

Welded plate
User and test manual

Manual scanning, display interpretation, examination preparation exercises and a working guide for Fabien.

Teaching application for a 245 × 250 × 10 mm welded plate with a symmetric double-V (X) preparation. Reference application: **P03**. All screenshots in this manual come from this version and its supplied scenario.

> **Start:** extract the entire package, open `OUVRIR_APPLICATION.html`, select English → Training → MWB → Witness pose. Follow the exercises in chapter 6.

This document supports functional testing and display interpretation. It is not a qualified inspection procedure or an approved certification course. The applicable scheme and assessment criteria must be agreed with the instructor.

5 November 2026 · FR/EN application and documentation · Fabien's camera-based interface, no headset.

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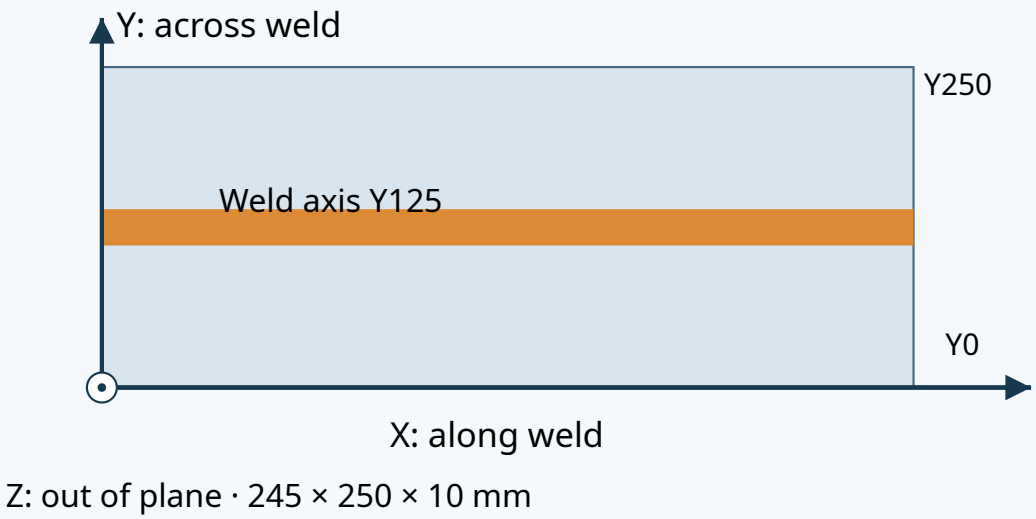
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1. Open the correct version

1. Extract the entire ZIP into a folder; do not work from the compressed-folder preview.
2. Open **OUVRIR_APPLICATION.html** in an up-to-date Chrome or Edge browser.
3. Check **prototype P03** in the heading and P03 in the local folder path. Close P01/P02 tabs to avoid comparing the wrong versions.
4. Select 🇬🇧 English. The offline application and manual require no installation or Internet connection.

The Streamlit version uses the same calculation and views in a web page. Deployment is described in chapter 12. PDF and HTML manuals are available above the application in Streamlit.

2. Specimen, axes and depth



Coordinate	Meaning
X: 0–245 mm	Longitudinal weld direction and main scan travel.
Y: 0–250 mm	Transverse direction. Weld axis: Y125 mm.
Global Z: 0–10 mm	Thickness, upward. TOP is Z10; BOTTOM is Z0.
Field/S-scan depth	Zero at the scanning face. TOP: depth = 10 − Z. BOTTOM: depth = Z.

Example: an inclusion at global Z3 mm is 7 mm deep when scanned from TOP. Do not confuse acoustic display depth and CAD global Z.

“X preparation” describes the shape in a Y–Z cross-section; it is not the longitudinal X coordinate. Four bevels are at 45°, the central land is 1.25 mm and the original 1 mm preparation gap is filled with weld metal. Top and bottom reinforcements are 18 mm wide and 2.5 mm high at the centre.

Weld toes are at Y116 and Y134 mm. HAZ bands are Y106–116 and Y134–144 mm. CAD partitions share boundaries without overlapping volumes; HAZ is not an air gap.

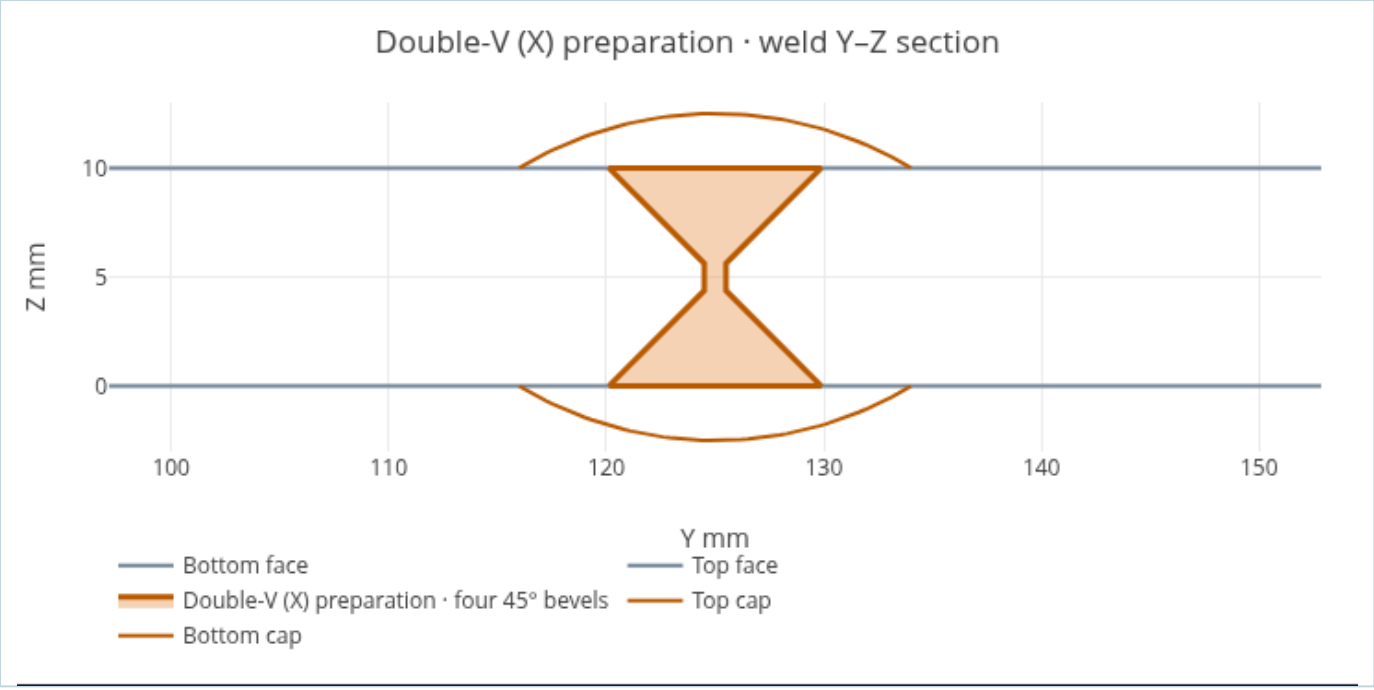


Fig. 1 — Symmetric double-V (X) preparation: four bevels and two caps. Global Z points upwards.

3. Select the examination method

Mode	Exercise purpose	Starting settings
SE 0° · Tx/Rx	Separate transmitting and receiving crystals. HAZ examination. Its witness pose targets an inclusion in the HAZ, not weld metal.	4 MHz L; 3 mm separation and 5 mm crossing depth are demonstration values.
MWB	Examine the weld from its sides using 45°, 60° or 70° shear waves.	4 MHz T; provisional 8 × 9 mm aperture and mechanical footprint.
PA	32 elements, 5 MHz; 35–70° sector and selected-angle A-scan.	Imasonic selected, 0.5 mm pitch, 12 mm elevation. Provisional 0.45 mm element width and 30 mm acoustic-path focus in steel.

The selected PA wedge concept gives a **nominal refracted 55° shear angle in steel**. This is not its physical wedge inclination. Exact model, internal geometry, velocity and hardware delay laws remain pending: P03 starts calculations at the steel interface.

Nominal A42-type carbon steel: VL5,920 m/s, VT3,240 m/s, density 7,850 kg/m³. Base plate, weld and HAZ share the initial velocities: **no artificial echoes are created at HAZ boundaries**. E7018 is an illustrative filler family, not a qualified welding procedure.

4. Controls and scanning gestures

Control	Action and what to observe
X, Y, heading	Steel emergence point and rotation about Z. Heading 90° points towards +Y; 270° towards −Y. Editing X/Y/heading calculates after the input is committed.
X−1/X+1; Y−1/Y+1	Manual 1 mm moves. Plan X−/X+/Y−/Y+ buttons and arrow keys use the local Step (mm) setting.
Drag in plan	Hold the left mouse button and drag in X/Y. A calculated coarse preview accompanies motion; the field grid refines on release. Click the plan to focus arrow-key controls.
TOP / BOTTOM	Select the scanned face. Depth zero moves to that face; specimen global coordinates remain unchanged.
Reverse heading	Add 180°. It changes beam direction; it does not transfer the wedge to the other weld side.
Calculate	Recalculate and add an acquisition. Use after changing angle, gain, frequency or probe settings.
Witness pose	Restore a known pose for the selected method and calculate. For MWB this also restores 45°; choose 60/70° afterwards when required.
Weld zoom / Full plate	Intentionally change the plan framing. Ordinary pose updates retain framing.
Reset views	Restore initial framing without clearing acquisitions. Manual zoom and 3D camera normally persist.
Stop / Reset acquisition	Stop travel and the current calculation, or clear accumulated poses/maps. Reset acquisition does not zero the probe pose.
Teaching truth	Show/hide known flaw markers. A truth marker is not a detection.

The green rectangle is the oriented wedge footprint including 1 mm clearance. The red dot is the emergence point and may be offset from the wedge centre. The orange plan band is the prohibited cap area.

Stops: dragging stops at caps and plate edges. Numeric entry can reposition the probe on the other side, as after lifting; sliding physically over a cap is not accepted as valid contact.

Intermediate positions are coalesced while calculations are busy. Free scanning guarantees neither video frame rate nor uniform spatial sampling. The **2 mm X increment belongs to raster scanning**. Wedge and Last calculation readouts distinguish current pose from the latest completed response.

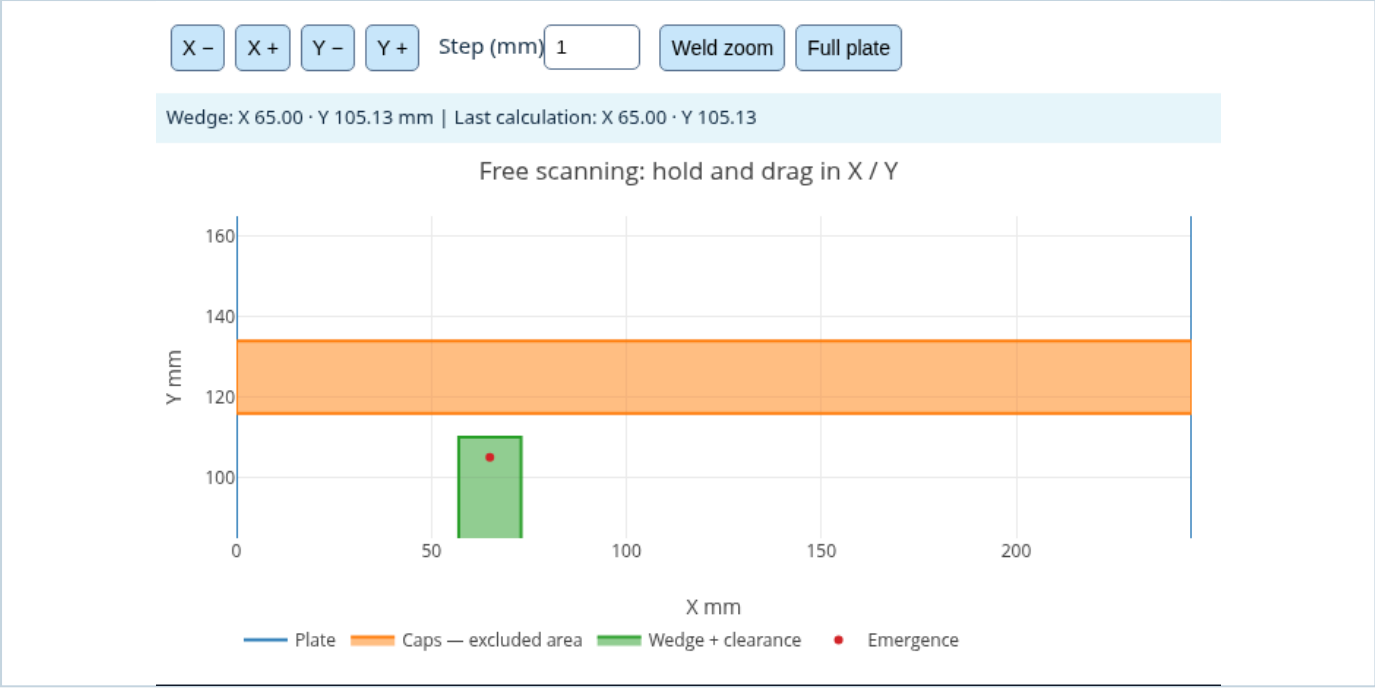


Fig. 2 — Scanning plan: green wedge, red emergence point, orange excluded band. Local buttons control X/Y.

5. What do the views mean?

5.1 CAD and central beam axis

The scene shows geometry, simplified wedge, emergence point and central-ray segments. Rotate with the mouse and zoom with the wheel. A ray is a geometric path: **it does not show the complete beam width**. Known flaw markers are teaching truth.

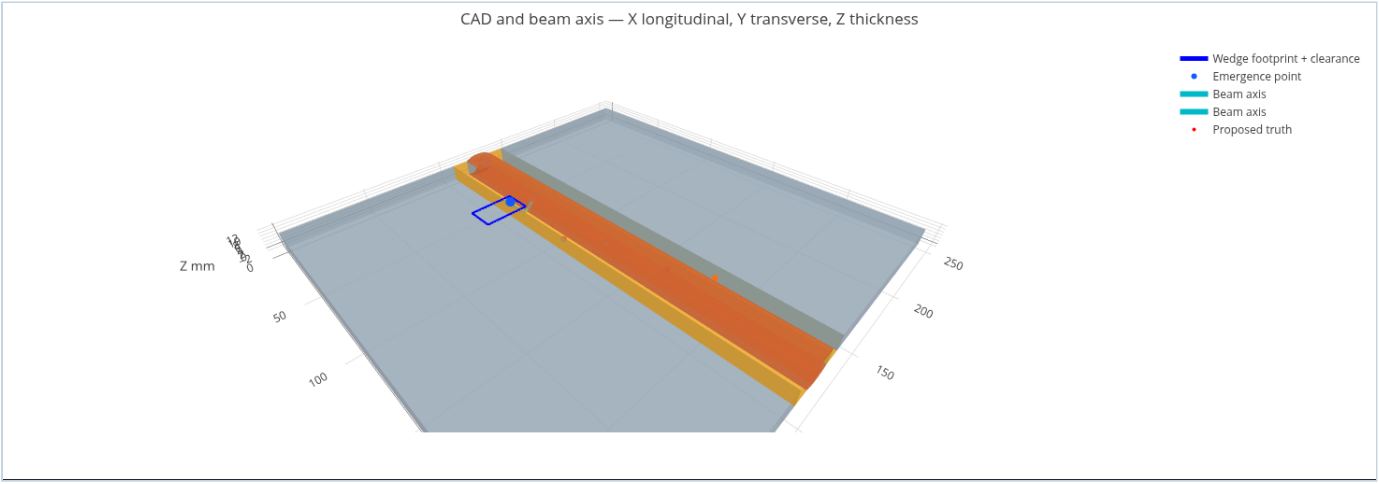


Fig. 3 — CAD and central ray at MWB45°, X65 / Y105.125 mm. Blue outline is the provisional footprint plus clearance.

5.2 A-scan: one pose and one angle

Horizontal axis: round-trip time in µs. Vertical axis: synthetic envelope, relative amplitude after gain, on a 0–1 scale. The title gives method, angle and calculated X/Y. This is not raw RF or amplitude calibrated against a regulatory reference block.

Motion can change travel time and amplitude. Gain does not change the path; it changes displayed response amplitude. Peaks above 1 are off-scale: reduce gain. Backwall echoes can saturate while gain is increased to explore a weak response.

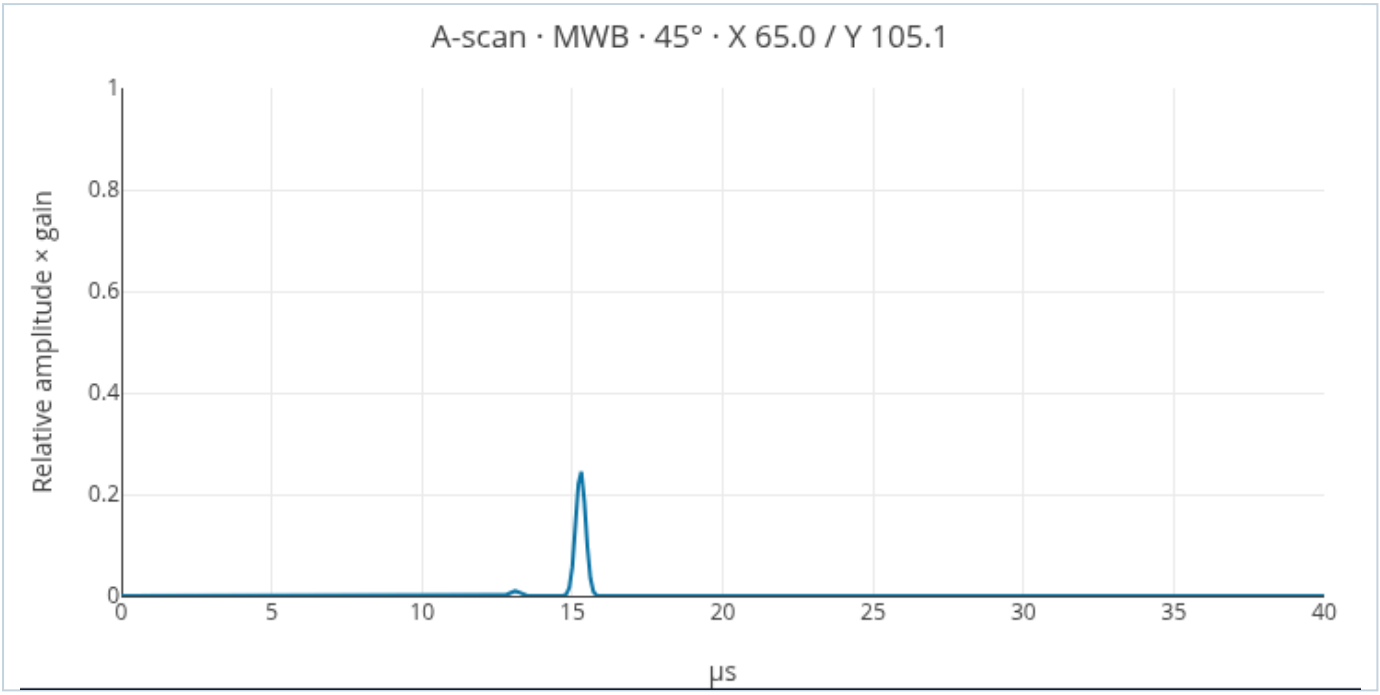


Fig. 4 — MWB45° witness pose,30 dB gain: main response near15.3 µs. Relative, uncalibrated amplitude.

5.3 X preparation and beam section

The joint view describes the specimen in global Y–Z. The adjacent section overlays the ray and, in training, flaws near the current X plane (±3 mm). A V-shaped backwall-reflected ray **does not mean a single-V weld preparation**: the joint remains double-V (X). At non-transverse headings the displayed ray is its Y–Z projection.

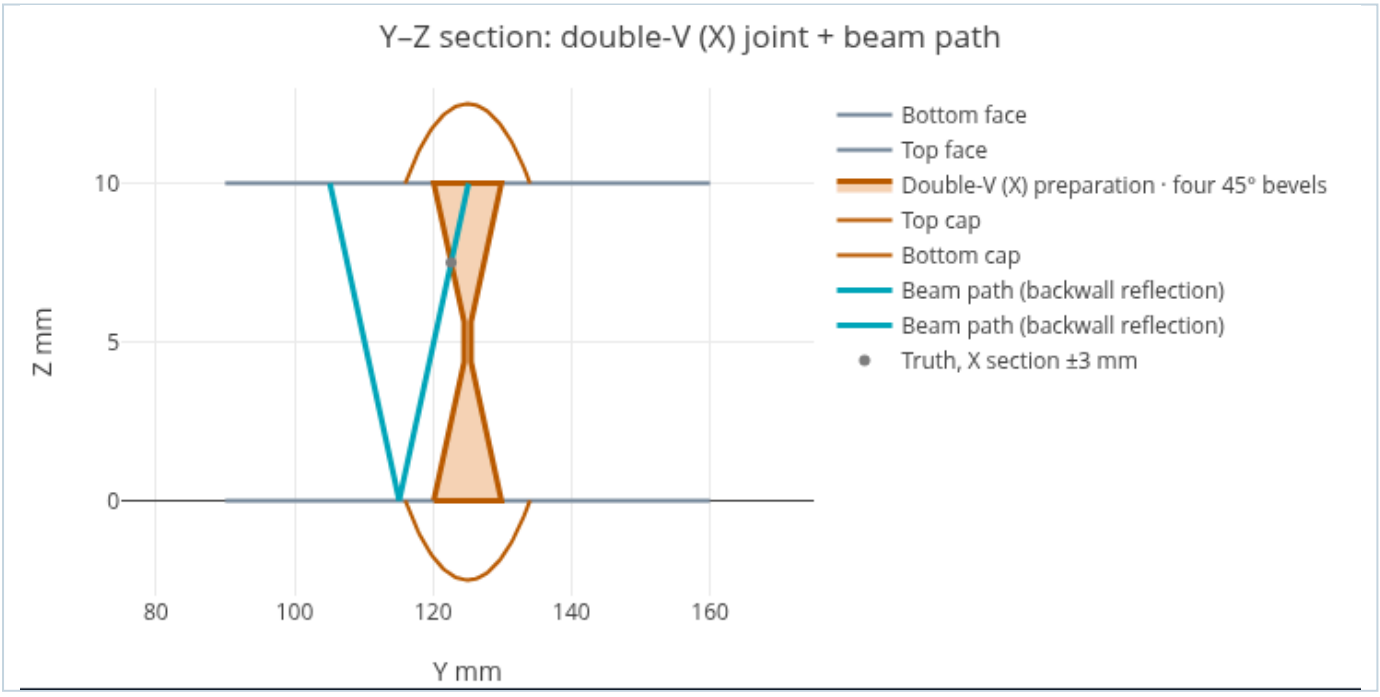


Fig. 5 — X preparation and reflected ray overlaid; the V-shaped ray does not define the joint preparation.

5.4 Incident and scattered fields

The first map shows harmonic incident amplitude including one opposite-face reflection. The second shows calculated scattering from active flaws, including return towards the probe. Complex contributions retain their phases. Maps are constant-X slices; white rays provide geometric guidance.

Depth increases from the scanning face. Initial Y window is fixed 70–180 mm. Motion preview uses a coarser grid than the released view, not a different sensitivity. Coloured fringes are interference in the harmonic model, **not numerous additional defects**.

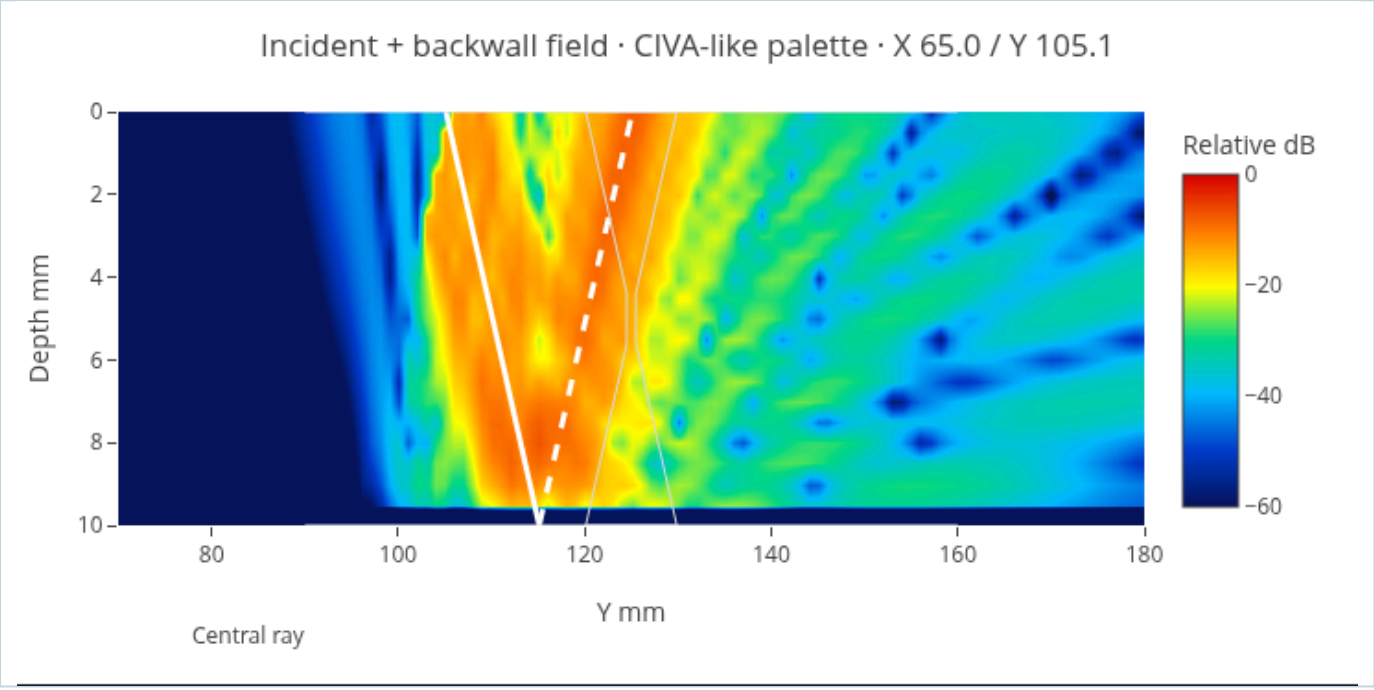


Fig. 6 — Incident and backwall field with white ray; depth zero at the scanned face.

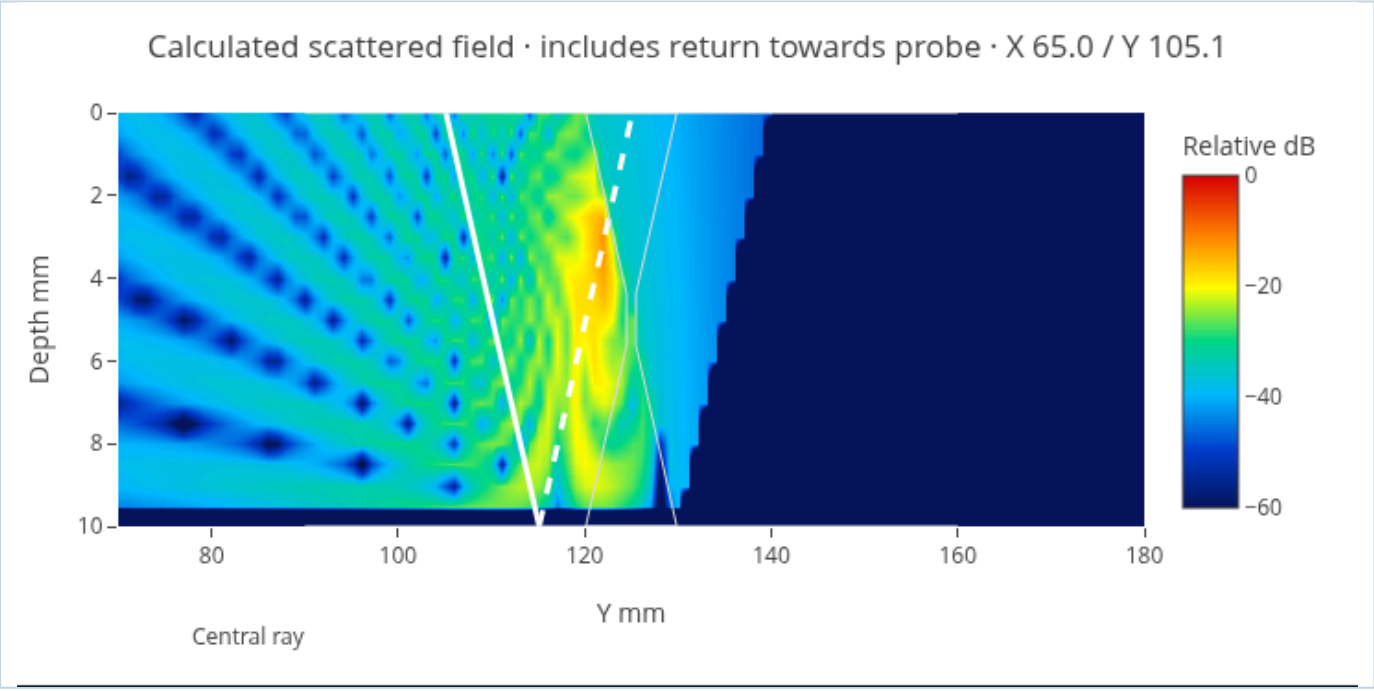


Fig. 7 — Calculated scattered field on the same plane and colour reference, including return towards the probe.

5.5 PA S-scan and single-element trace

PA sector 35–70° is backprojected into Y–depth. A coloured point represents a time-angle response placed along an assumed ray, not a pixel of true flaw geometry. Angular Δθ in degrees is separate from travel increment in millimetres. The selected A-scan angle (default 52°) is calculated separately and may lie between sampled sector angles.

For MWB/SE this becomes a single-ray Y–depth trace; **it is not a PA sector scan**.

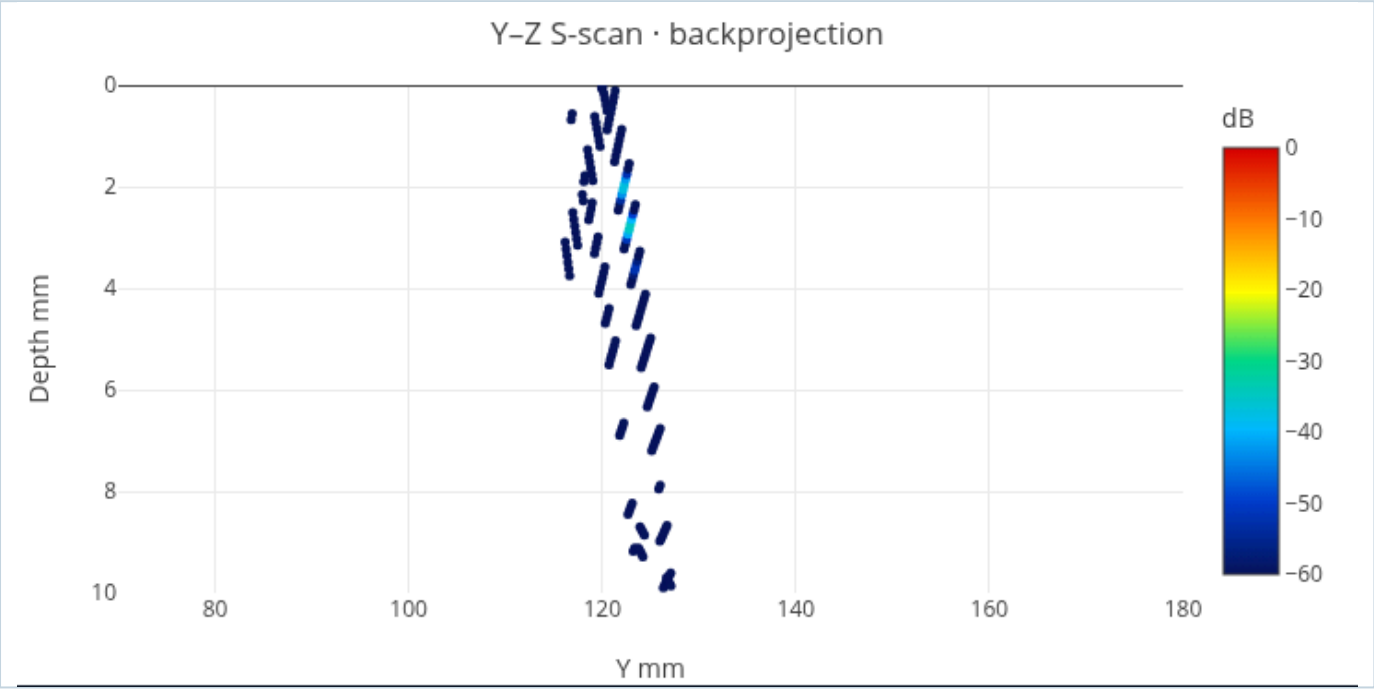


Fig. 8 — PA example:35–70° sector, X65 / Y100 mm, depth measured from TOP.

5.6 B-scan: acquisition history

Horizontal axis: **recorded pose number**, not X in millimetres. Vertical axis: time in µs, increasing downwards. Each column contains that pose's responses. In PA, P03 uses the **first sector ray,35°**, rather than the selected A-scan angle.

Curves can arise from changing path length as Y or heading changes. One flaw can generate several paths. Lines are therefore not flaw outlines. For comparisons, clear acquisitions and repeat a defined trajectory rather than mixing arbitrary angles and motions.

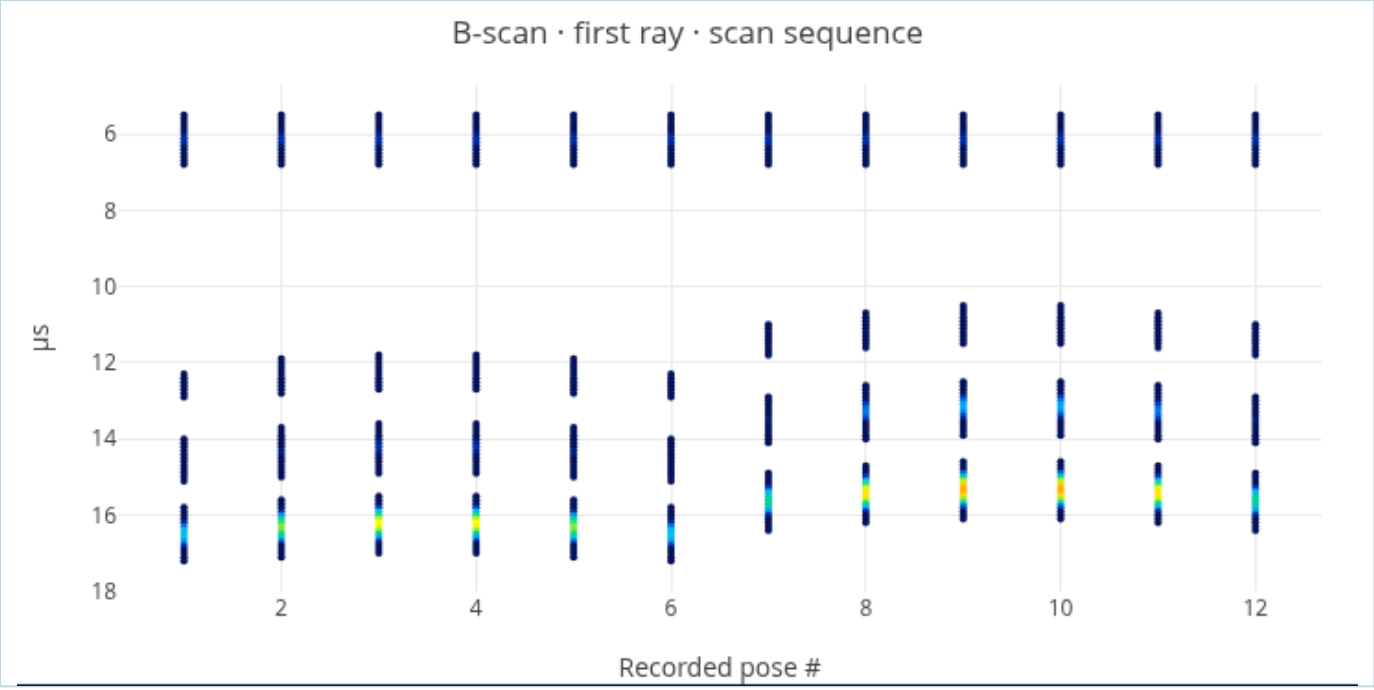


Fig. 9 — Exercise raster:12 positions numbered horizontally; time in µs.

5.7 3D accumulation: backprojected responses

Axes: specimen global X/Y/Z. Colours encode relative response after gain. Recorded events are placed on assumed paths using time and angle. One flaw can create a cloud and multiple path images. **This is not exact flaw geometry, validated tomography or automatic flaw-volume measurement.**

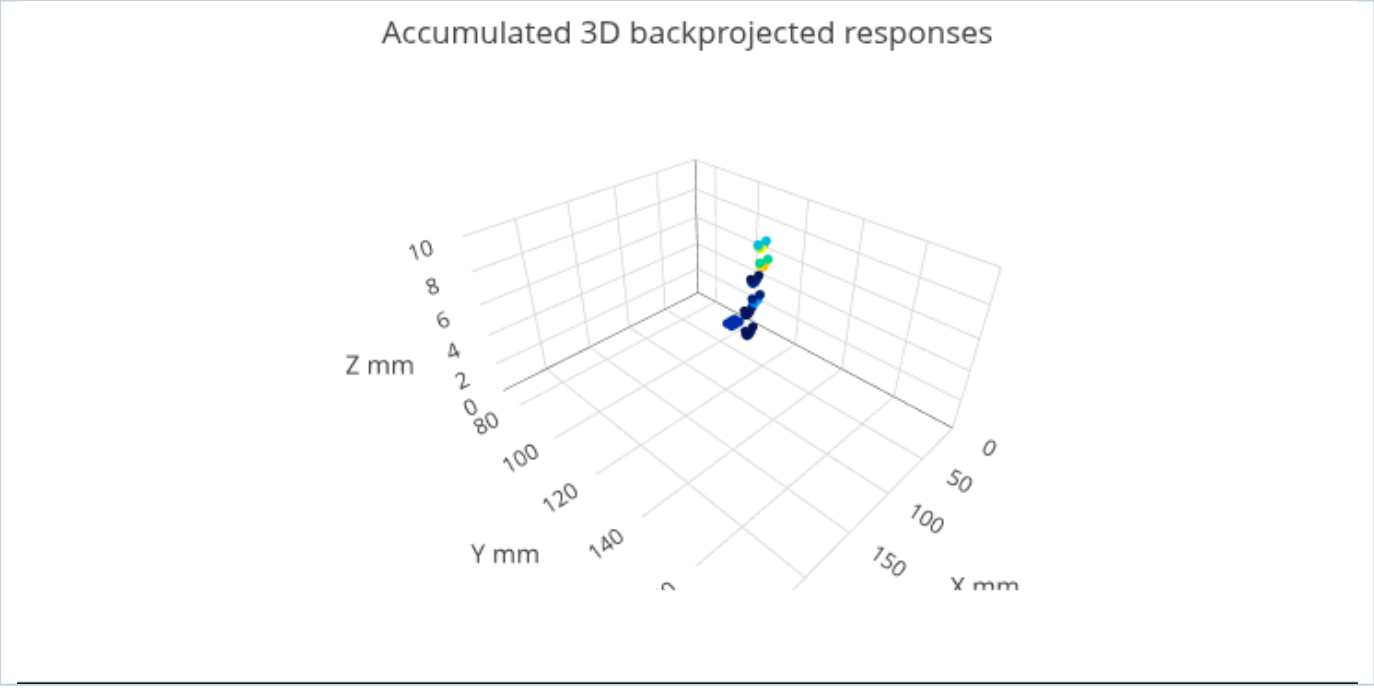


Fig. 10 — Accumulated responses from the same exercise: global X/Y/Z backprojection, not true flaw geometry.

6. Recommended test sequence

Begin in Training, truth enabled, TOP face. Record pose, angle, gain and observations after each exercise. Values describe the proposed scenario, not measurements of a manufactured specimen.

A — First lack of fusion with MWB45°

1. Select MWB → Witness pose. Check X65, Y105.125, heading 90°, TOP, MWB45°.
2. Gain 30 dB → Calculate. The model predicts a reflected MF1 path near **15.3 µs**. Amplitude depends on pose and model; this is not an acceptance threshold.
3. Use Y−1 then Y+1. Compare time, amplitude and fields on unchanged scales.
4. Move several X−1/X+1 steps and return to X65. Observe response loss away from the reflector.
5. For 60°/70° comparison, restore the witness pose, then change the angle and Calculate. A common position need not be optimal for all angles.

B — HAZ inclusion using SE

1. SE → Witness pose: X105, Y111, heading 90°, TOP.
2. Gain 0 → Calculate to view the backwall without major clipping; increase gain to explore the weak response.
3. INC1 is at global Z3, hence depth 7 mm. Direct geometric time is about 2.42 µs; backwall is about 3.42 µs with the proposed dual-crystal geometry.
4. This inclusion is in the left HAZ. The exercise does not establish that SE detects weld lack of fusion.

C — 32-element PA and selected angle

1. PA → Witness pose: X65, Y100, heading 90°, TOP.
2. Check 35–70°, Δθ2° and selected A-scan 52°. Calculate.
3. Read S-scan and selected-ray A-scan. Change selected angle and Calculate; B-scan remains based on 35°.
4. Move Y in 1 mm steps and check stable framing and scales.

D — Reproducible raster

MWB45°, TOP, heading 90°: Xmin 60, Xmax 70, Ymin 103, Ymax 105, ΔY2. Reset acquisition, then Start raster. **Expected:12 poses** — six X values 2 mm apart on each of two Y lines, with reverse X travel on the second line. This exercise is checked in P03.

The default PA rectangle X50–80, Y95–100, ΔY5 gives 32 poses after clearing acquisition. Wedge contact must remain valid.

E — Examine from the opposite side

For MWB45°, enter X165, Y144.875, heading 270°, TOP, then Calculate. This pose targets MF2 from the right using a reflected path. Intermediate input combinations may briefly warn about contact: finish all values and check Valid contact. Witness pose returns to the left.

7. Colours, gain and framing



Relative amplitude: low → high. 20 log10(A/Aref).

The palette progresses from dark blue through blue/cyan, green, yellow and orange to red. It resembles familiar CIVA and MANTIS/GEKKO displays but **is not an exact verified manufacturer LUT**. Comparisons require matching thresholds, units and references as well as colours.

Value	Amplitude relative to reference
0 dB	1
−20 dB	0.1
−40 dB	0.01
−60 dB	0.001

Fields share a fixed scalar reference 0.05 in propagator units, displayed over−60 to 0 dB. They are not normalized independently at each pose. Warm colours mean high calculated amplitude, not certain flaw presence or weld rejection.

Gain affects A-scan, S-scan/trace, B-scan and 3D responses, not the two fields' fixed reference. These are not calibrated pressure maps. Keep gain unchanged during comparisons and record any adjustment.

Motion preserves zoom and 3D camera. Reset views if a beam or point moves outside the frame. Weld zoom/Full plate changes only the plan.

8. Training, exam preparation and report

In Training, use known markers to understand paths. Exam preparation hides/disables the truth table, scenario import and witness pose, and clears prior acquisitions. Session export omits response truth identifiers and exact truth coordinates.

P03 remains a local exercise, not a secure examination platform. Source and scenario are in the package, and teaching fields remain available. Required masking for a formal test must be specified by the instructor.

1. Identify the specimen, method and accessible surfaces.
2. Agree the certification scheme, level and procedure with the instructor.
3. Plan real-hardware verification of velocity, zero, emergence, angle and sensitivity on reference blocks; P03 does not qualify these settings.
4. Define trajectories from both sides, angles and uncovered areas.
5. Record indications, locations and uncertainties; distinguish geometry echoes and flaws.

6. Complete Operator, Scheme/procedure and observations. Export the readable HTML report and the JSON session for traceability.

Open the report in a browser and print to PDF if needed. Full characterization and an acceptance decision are not generated automatically. P03 has no selected certification reference, so no normative acceptance criteria are applied.

9. Flaws and ML variation preparation

ID	Type	L × W / Ø mm	X	Y	Z	Rx / Ry / Rz °
MF1	Lack of fusion	2 × 0.5	65	122.625	7.5	-45 / 0 / 0
MF2	Lack of fusion	4 × 0.5	165	127.375	7.5	45 / 0 / 0
INC1	Spherical inclusion	Ø 0.5	105	111	3	0 / 0 / 0
INC2	Spherical inclusion	Ø 0.5	185	139	7	0 / 0 / 0
FIS1	Sawtooth crack	2 × 0.5	125	125	3	90 / 0 / 0

Locations form a proposed teaching scenario. Lack-of-fusion and crack objects are non-cutting reference surfaces; air-layer opening is undefined. The two spheres have assumed 0.5 mm diameter.

In Instructor scenario, export INDICATIONS_ML.json, edit a copy and reimport it. Preserve canonical keys and decimal points. Fields include dimensions, X/Y/Z and Rx/Ry/Rz in degrees with R = Rz Ry Rx. These are flaw rotations, not probe angles.

P03 import bounds: up to 20 rows; planar length>0 and≤10 mm, width 0.1–3 mm; sphere diameter>0 and≤3 mm; centres inside the plate. Software bounds are not physical validity domains. Import changes the acoustic scenario, not the STEP solid geometry.

For ML, separate ground truth, scenario, probe settings and acquisition. Validate variation domains with the instructor and split training/evaluation by specimen/scenario. P03 contains neither a trained network nor validated mass data generation.

10. Fabien’s working guide — camera, no headset

The goal is probe control using a PC or external camera without a headset. P03 supplies a pose interface and reference scene. Video detection, camera/plate calibration and marker tracking must connect to Fabien’s system.

1. Start with the same MWB45° exercise as the student and the FR or EN manual.
2. Open the Fabien interface panel and choose Fabien camera.
3. Test the sample JSON already calibrated in plate coordinates. Position refers to the steel emergence point, not the video marker centre.
4. Check low-confidence, lift/tilt, edge and cap rejection. Tracking loss must not become valid acoustic contact.
5. Connect the camera adapter to the JavaScript API and test accuracy, latency and acquisition rate.

```
 await window.ByteNDTPlate.applyTrackedPose({
   sequence: 1, timestamp_ms: 0,
   position_mm: [105, 111, 10],
   euler_xyz_deg: [0, 0, 90],
   face: 'TOP', confidence: 0.95
 });
```

FABIEN_CAMERA contains the complete contract and sample JSON. Positions are mm, conversion m = mm ×0.001; rotations Rz Ry Rx. TOP: [0,0,heading] at Z10; BOTTOM: [180,0,heading] at Z0. P03 accepts confidence≥0.7, contact Z tolerance 0.5 mm and tilt tolerance 3°, then constrains the acoustic pose to planar contact. The `byte-ndt-plate-response` event supplies the latest accepted acquisition. P03 performs no video capture.

Exported sessions contain 6D poses, profiles, synthetic responses and contact states, not video or a finished immersive client. Await each call or explicitly handle cancellation of older calculations. No camera frame rate is guaranteed.

11. Troubleshooting

Symptom	Check
No visible echo	Check valid contact, method, angle, side, heading and X near the flaw. Restore Witness pose. A central ray grazing a flaw need not produce a strong echo.
Off-scale warning	Reduce gain and Calculate. This is not a software error.
Gain does not change fields	Expected: fields retain a fixed reference. Gain affects response displays.
WELD_TOE / PLATE_EDGE	Wedge plus clearance reaches a cap or edge. Move back or reposition on the opposite side with the correct heading.
Large 3D point cloud	Many poses, angles and paths have accumulated. Do not interpret it as flaw size; clear and repeat a known trajectory.
Uncertain version	Check prototype P03 and the file path; close P02. P03 has two separate coloured field maps.
Coarse field while moving	Calculation preview. Full grid appears on release. Compare completed poses when assessing amplitude.
Camera does not turn on	Expected before integration: P03 does not read video. Fabien must connect tracking and calibration.
Blank page or 3D view	Reopen in Chrome/Edge and check WebGL. Allow the initial calculation to finish; recover the complete package if source files are missing.

12. GitHub and Streamlit launch

P03 is published on a dedicated repository branch. To create a web service, **create a new application** from the owner’s Streamlit Community Cloud account. Do not replace the existing Nozzle service.

Dépôt / Repository	`jmpuybouffat/01_Paut_Training_-`
Branche / Branch	`codex/tole-x-cao`
Fichier principal / Main file	`plate_app.py`

GitHub · codex/tole-x-cao

1. Sign in to Streamlit Community Cloud using an account with repository access.
2. Choose Create app from GitHub and enter the three fields above.
3. If offered, select Python 3.12 under advanced settings; it is locally tested. Runtime dependencies are in root requirements.txt.
4. Select Deploy, wait for startup and check FR/EN PDF buttons and MWB45° Witness pose.
5. Record the actual generated URL and share it with the chosen reviewers. This manual does not presume that a public plate Streamlit URL has already been created.

Offline HTML remains server-free. Streamlit requires Internet access and owner-account authorization. GitHub sources and the package can be handed to Fabien; do not commit named student reports.

13. Checked behaviour and remaining validation

Software checks cover X/Y motion, stops, increments, languages, exports, camera poses, framing and simple analytic travel times. Double-V CAD has been checked. Manual figures are actual P03 outputs.

Acoustics remain central-frequency scalar: finite aperture, planar surface approximation, weak Born inclusions and one opposite-face image per transmit/receive path. Pulses are synthetic. No wedge propagation, mode conversion, GTD, cap scattering or full elastic solver. Received backwall echoes are included for SE/MWB, not PA in P03.

Operators and instructor must still validate gestures, exact hardware, sensitivity, coverage, scenarios, terminology, scheme criteria and comparisons with measurements or reference software. Similar colours to CIVA/MANTIS/GEKKO do not validate the acoustic engine.

For useful feedback, provide a screenshot plus P03, language, method, face, X/Y, heading, angle, gain, action, observed result and expected result. Change one parameter at a time during comparative tests.